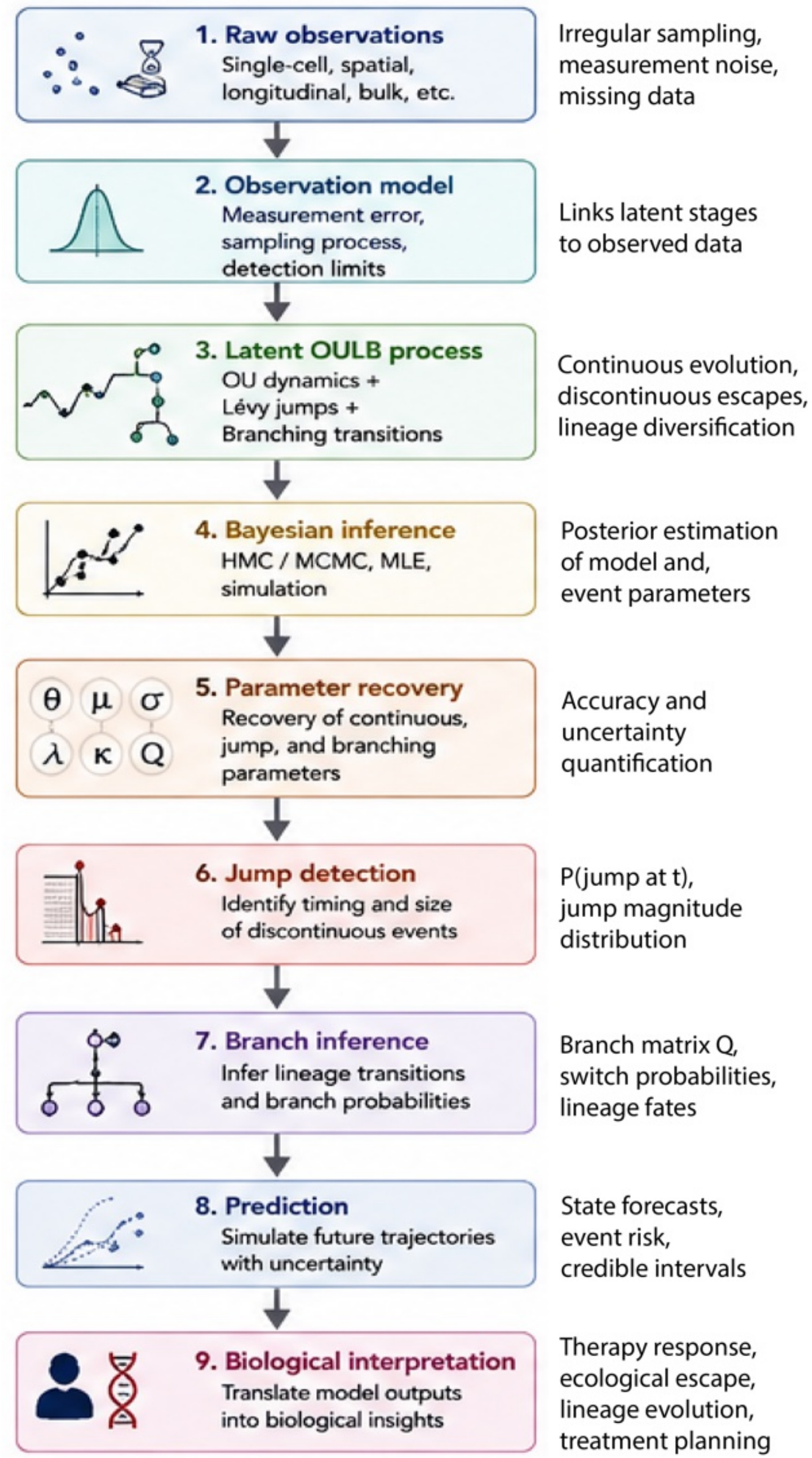
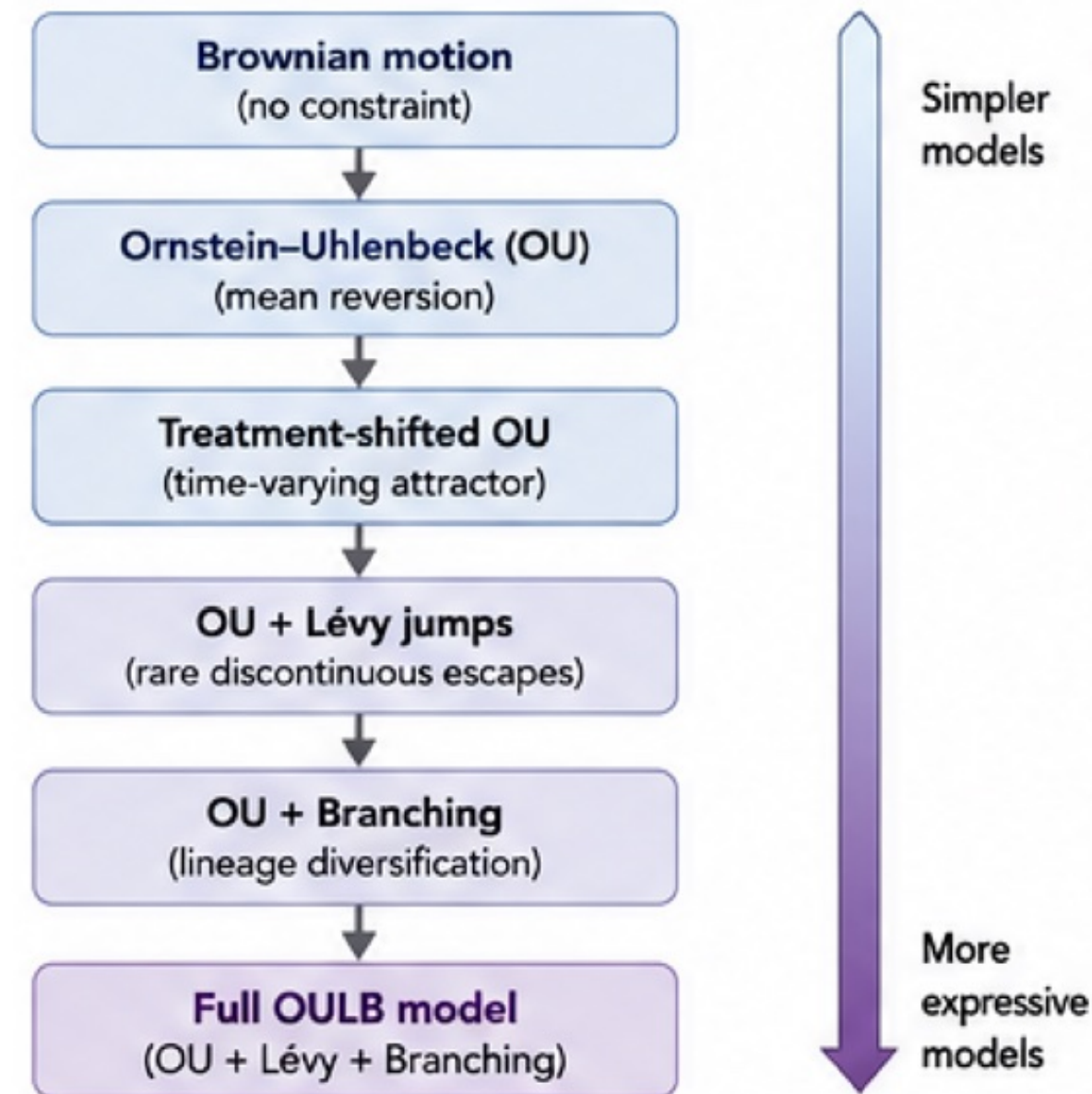


Unified computational workflow of the Ornstein-Uhlenbeck-Lévy-Branching (OULB) framework

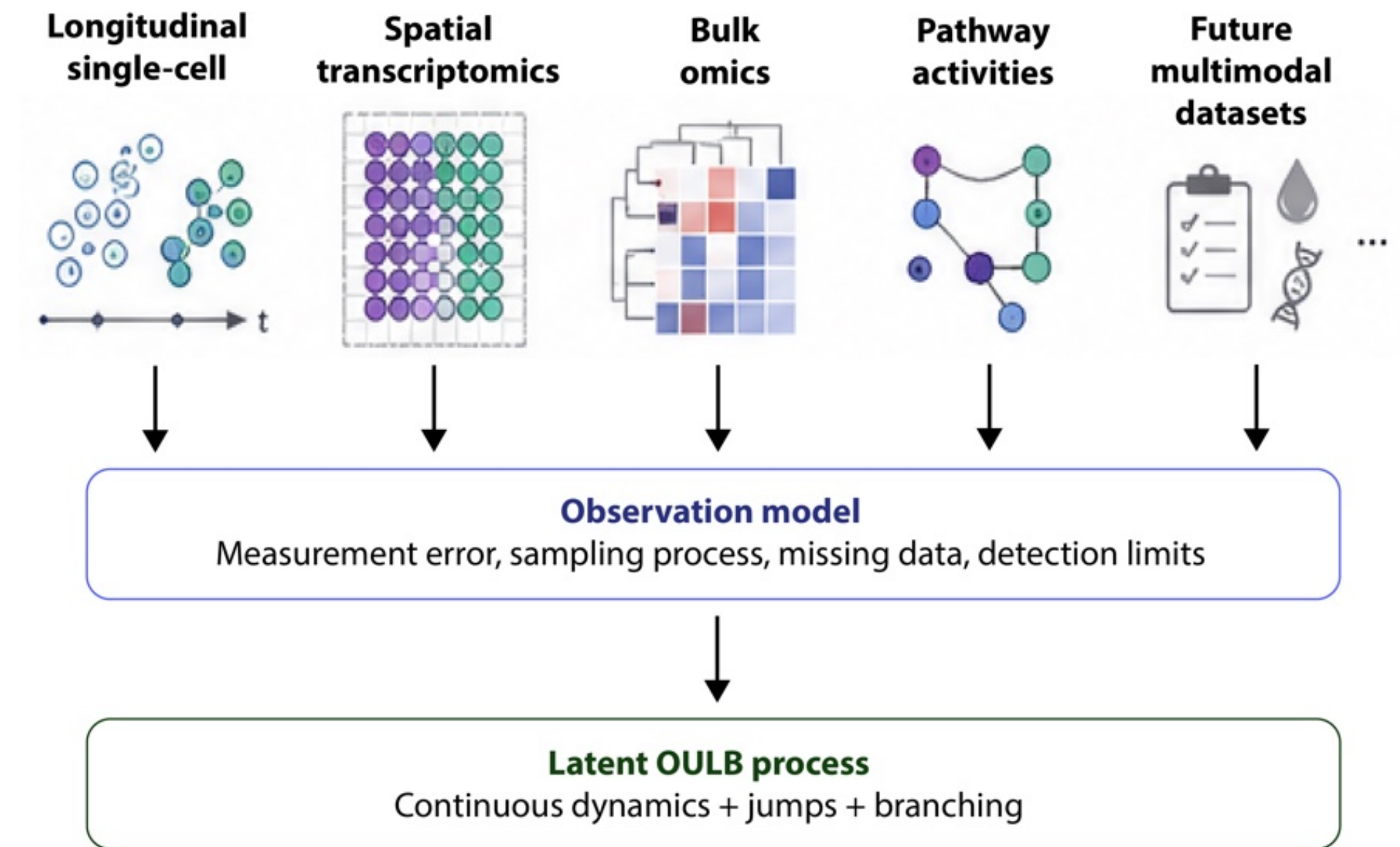
A End-to-end computational workflow



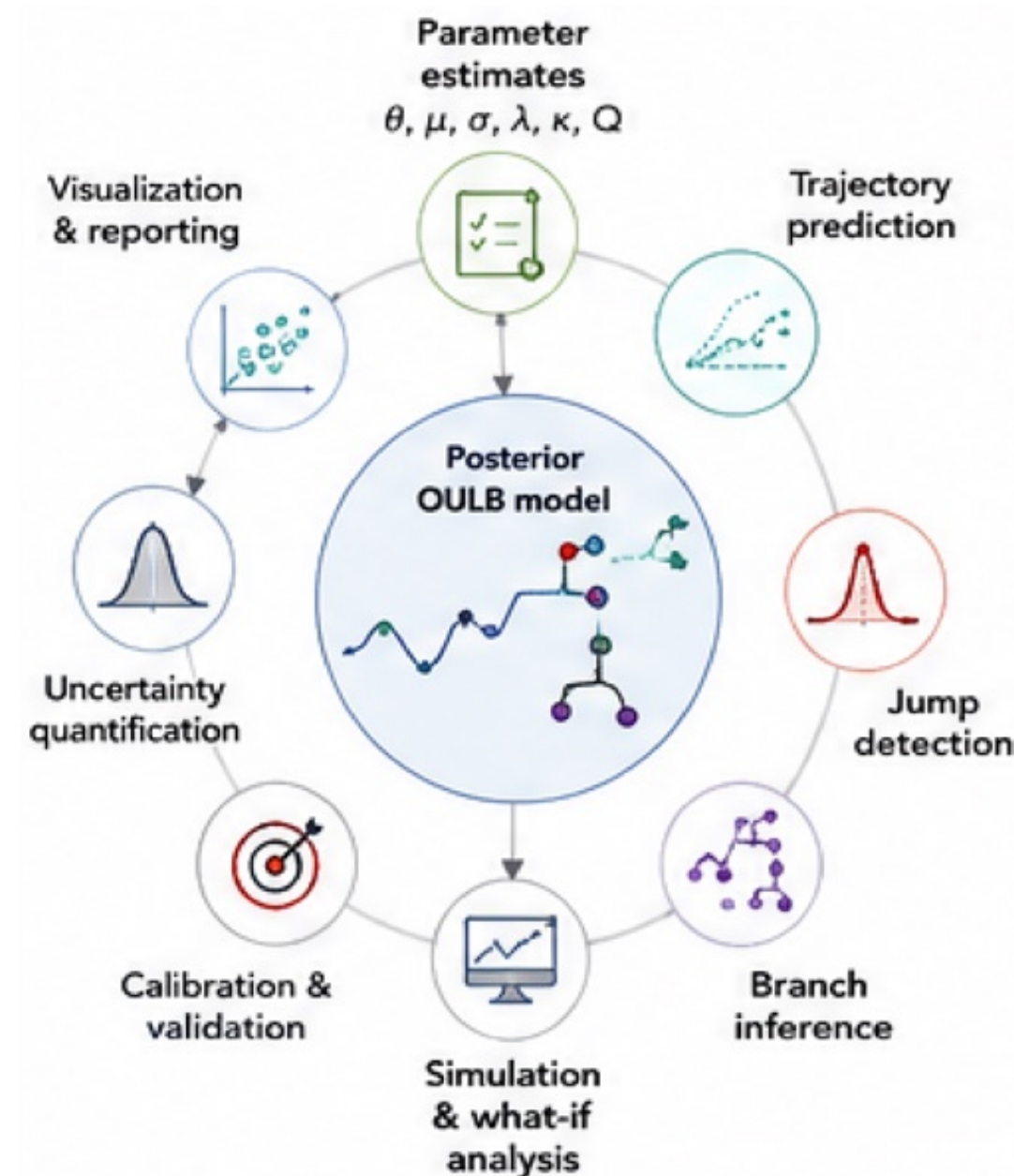
B Nested model hierarchy (special cases)



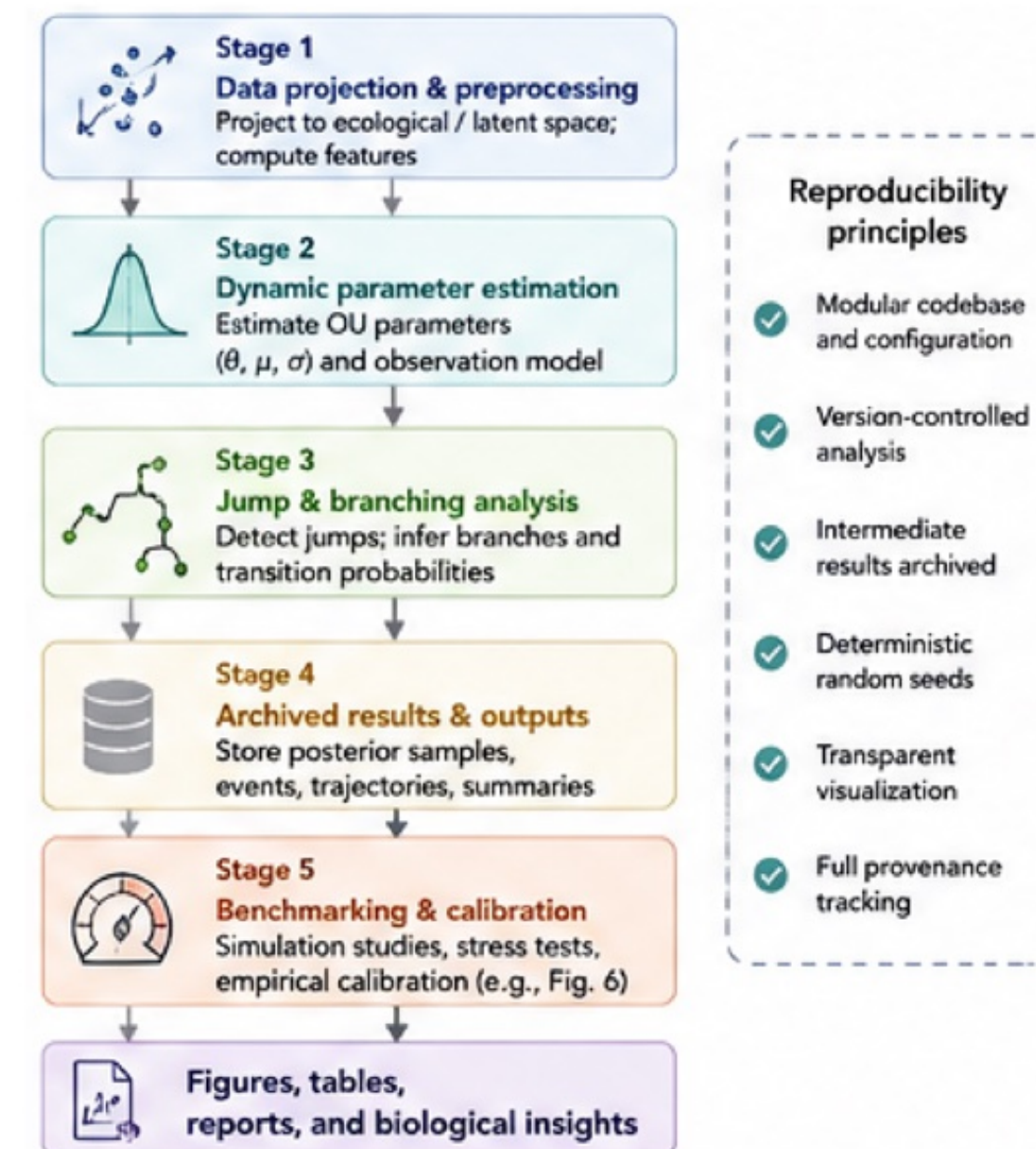
C Broad data modalities supported



D Outputs of the unified framework



E Software architecture & reproducible pipeline



F Take-home message

The OULB framework integrates stochastic dynamics, discontinuous evolutionary events, lineage diversification, observation processes, Bayesian inference, empirical calibration, and reproducible benchmarking into a single interpretable computational framework for studying cancer evolution.

- Unified mathematical foundation
- Flexible across data types and biological contexts
- Validated through simulation and real-world calibration
- Enabling interpretable insights and clinical relevance